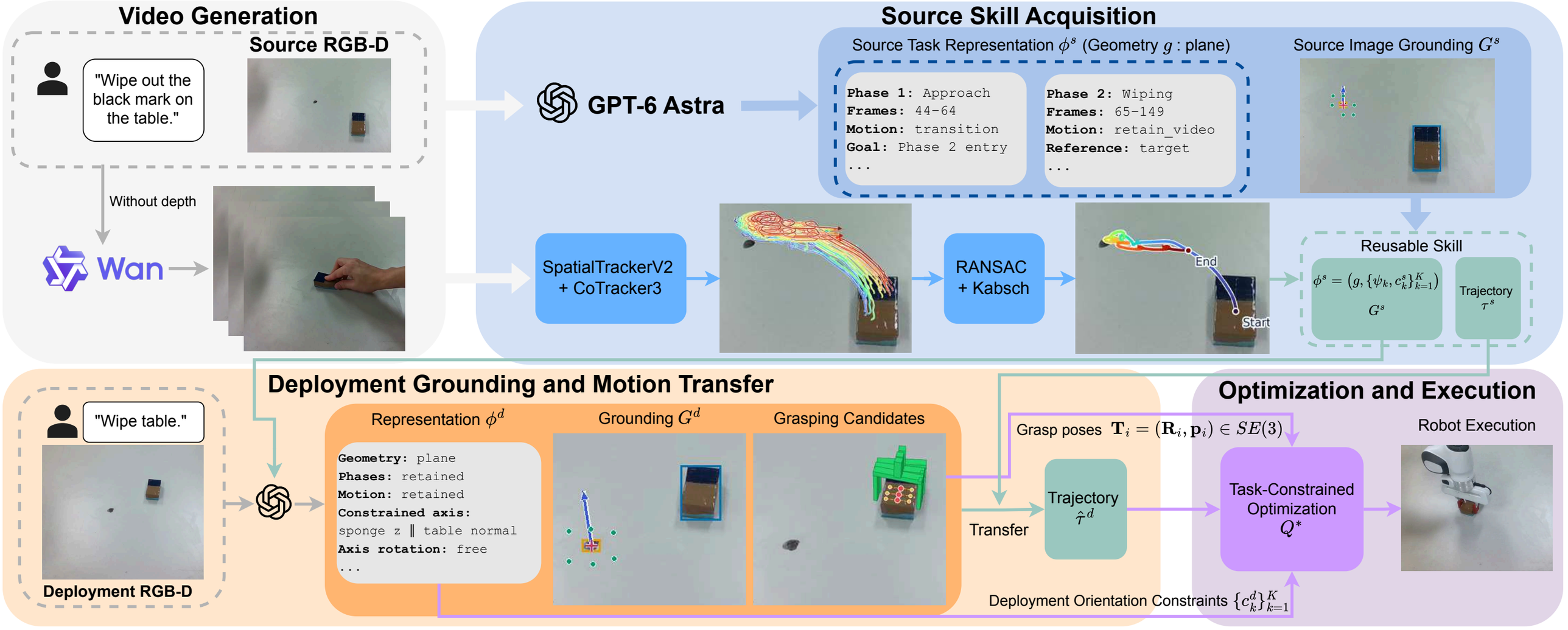

Video Generation
Source RGB-D
"Wipe out the black mark on the table."
Without depth
Wan
Source Skill Acquisition
GPT-6 Astra
Source Task Representation ϕ^s (Geometry g : plane)
Phase 1: Approach
Frames: 44-64
Motion: transition
Goal: Phase 2 entry
...
Phase 2: Wiping
Frames: 65-149
Motion: retain_video
Reference: target
...
Source Image Grounding G^s
SpatialTrackerV2 + CoTracker3
RANSAC + Kabsch
End
Start
Reusable Skill
$\phi^s = (g, \{\psi_k, c_k^s\}_{k=1}^K)$
G^s
Trajectory τ^s
Deployment Grounding and Motion Transfer
"Wipe table."
Deployment RGB-D
Representation ϕ^d
Geometry: plane
Phases: retained
Motion: retained
Constrained axis:
sponge z ∥ table normal
Axis rotation: free
...
Grounding G^d
Grasping Candidates
Transfer
Trajectory $\hat{\tau}^d$
Optimization and Execution
Grasp poses $\mathbf{T}_i = (\mathbf{R}_i, \mathbf{p}_i) \in SE(3)$
Task-Constrained Optimization Q^*
Robot Execution
Deployment Orientation Constraints $\{c_k^d\}_{k=1}^K$